

An improved transmission efficiency prediction method for nonlinear characteristics of the cycloid reducer

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Abstract

This study aims to accurately predict the nonlinear characteristics of transmission efficiency of cycloid reducers under different operating conditions. Firstly, equivalent modeling of the multi-source errors (MSEs) in the designed cycloid reducer is conducted. Force analysis algorithms considering MSEs are proposed for the cycloid drive mechanism, the output mechanism, and the bearings. Secondly, mathematical models are established for the load-dependent power losses, while an equivalent test is used for modeling load-independent power losses. Subsequently, an improved transmission efficiency prediction (TEP) method for cycloid reducers is proposed, which is then applied to the performance prediction of a prototype under different operating conditions. The advantages of the improved TEP method over the conventional method are discussed, and the influences of MSEs and load-independent power losses on the nonlinear characteristics of transmission efficiency are summarized. Finally, tests are conducted for the reducer prototype, and the test results are found to be in good agreement with the results obtained by the proposed TEP method. The main contribution of this study is to establish a solid algorithmic and modeling foundation for the optimal design of nonlinear transmission efficiency in cycloid reducers and provide reliable guidance for their engineering applications.

Keywords

Nonlinear characteristics, transmission efficiency, cycloid reducers, multi-source errors, load-independent power loss, TEP method

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Introduction

Transmission efficiency,¹ as a key performance indicator of the gear unit, such as cycloid reducers, directly affects the effective utilization of energy. Meanwhile, due to multi-source errors (MSEs) like manufacturing and assembly errors,² the actual transmission efficiency of the prototype differs from ideal conditions, complicating accurate cycloid reducer efficiency prediction. Therefore, the study of the transmission efficiency prediction model of cycloid reducer benefits both theoretical understandings of the reducer's energy conversion and practical design optimization, enhancing energy efficiency and guiding engineering practices.

For decades, the transmission efficiency analysis model of the cycloid reducer has been refined to achieve accurate predictions during the design phase. Malhotra and Parameswaran³ established a force analysis model for cycloid reducers under ideal conditions, exploring the impact of varying friction

coefficients on transmission efficiency. Li et al.⁴ designed a double crank ring plate cycloid reducer, achieving over 90% efficiency in rated conditions. Sheu et al.⁵ studied the influence of design parameters on the transmission efficiency of a cycloid reducer with a pin-type output mechanism. Gorla et al.⁶ proposed a simplified analysis for cycloid gear meshing, power loss, and transmission efficiency, validated by tests. Blagojevic et al.⁷ emphasized the non-negligible effect of friction coefficients on the transmission efficiency of the cycloid drive mechanism, especially for

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small gear ratios. Mackic et al.⁸ found that cycloid drive mechanisms with different design parameters often exceed 90% efficiency. Sensinger⁹ investigated four cycloid reducer cases, concluding that designs with rollers have the highest efficiency. Hsieh et al.¹⁰ studied the transmission performances of cycloid drives with and without pins based on the FEM and observed higher efficiency in toothless cycloid reducers, contradicting Sensinger's⁹ findings. Mihailidis et al.¹¹ incorporated EHL theory into their cycloid reducer efficiency model, noting minimal nonlinear friction effects below a 0.05 coefficient. Sun et al.¹² studied the starved lubrication performance of cycloid drives using a numerical thermo-elastohydrodynamic model, optimizing oil film thickness under different conditions. Wang et al.¹³ presented a genetic algorithm-based optimization for the cycloid reducer design, aiming for minimal volume and maximal efficiency. Blagojevic et al.¹⁴ presented the numerical analysis of the stress state of the cycloid disc in the cycloid reducer, and the analysis results are in good agreement with the results obtained by the experimental method. Blagojevic et al.¹⁵ examined the efficiency of plastic cycloid reducers, noting significant discrepancies between theory and tests at low speeds but good agreement at higher speeds. Pham and Ahn¹⁶ modeled the efficiency of a cycloid reducer, accounting for tolerances. They found efficiency increased with load but decreased with larger tolerances, though lacking prototypes for validation. Artyomov et al.¹⁷ examined the forces in cycloid drives with backlash, showing increased forces and reduced sliding friction with greater backlash, contrary to Pham's findings.¹⁶ Jian and Deyu¹⁸ proposed a design method for cycloid tooth profiles, focusing on meshing efficiency for theoretical improvements. Wang et al.¹⁹ examined the impact of various cycloid gear profile modifications on transmission efficiency. Wang et al.²⁰ developed a transmission efficiency model for modified cycloid gears in RV reducers through numerical analysis of mixed EHL. Dion et al.²¹ dynamically modeled cycloid reducer efficiency, noting agreement with tests only when considering nonlinear contact stiffness in multi-tooth meshing. Vasić et al.²² compared two main efficiency models for cycloid reducers, revealing differences in power loss calculations and contact considerations. Matejic et al.²³ compared the size and efficiency of conventional and new two-stage cycloid reducers, finding the new design more compact and efficient. Jang et al.²⁴ investigated the influence of cycloid tooth profile design parameters on the meshing pressure angle, performing load-bearing contact analysis considering friction, and guiding efficient cycloid reducer design. Wang and Mao²⁵ optimized RV reducer design parameters for minimal power loss and volume, validating the approach through examples for high-power-density designs. Jiang et al.²⁶ considered the nonlinear effects of assembly dimension chains on cycloid drive tooth backlash in their

efficiency model, finding higher efficiency with needle roller cages due to reduced friction. Pabiszczak and Kowal²⁷ designed an externally driven cycloid transmission and analyzed its frictional power losses, showing the highest losses at the high-speed bearing, several times those of the cycloid drive. Xu et al.²⁸ modified a cycloid reducer with a rotatable output pin mechanism, achieving an efficiency of 83.39% under rated conditions in tests. Matejic et al.²⁹ conducted a dynamic analysis on the stress-strain state of a cycloid disc within a cycloid reducer through experimental and numerical methods, with both experimental and simulation results showing a deviation between 3% and 15%.

Based on previous theoretical and experimental research, it is evident that the nonlinearity of transmission efficiency in cycloid reducers is significantly influenced by speed and torque conditions. However, the reasons for this nonlinear phenomenon have not been adequately reported. Xu et al.'s^{30,31} studies indicate that the friction torque, unrelated to internal load in the reducer, varies nonlinearly with speed changes, an aspect not fully considered in existing transmission efficiency models. This is especially problematic in low-power reducers where nonlinear friction losses can affect efficiency predictions. Moreover, models analyzing transmission errors^{32–36} in cycloid reducers have demonstrated the non-negligible impact of MSEs on transmission characteristics. The remaining minority of analytical models for transmission efficiency that consider the influence of actual manufacturing factors also simplify the modeling of MSEs. This results in inaccurate force analysis of critical components during the process, making it difficult to theoretically identify nonlinear phenomena of the transmission efficiency of reducers. Furthermore, the transmission efficiency of a reducer under specific operating conditions is typically measured as an average of transient efficiencies over some time. This approach can obscure nonlinear phenomena caused by MSEs under constant operating conditions. Therefore, there is a need for further investigation into the impact of MSEs and load-independent power consumption on the nonlinear characteristics of cycloid reducer transmission efficiency.

This study aims to delve deeper into the underlying mechanisms of nonlinear transmission efficiency in cycloid reducers under various operating conditions and develop a precise prediction method, which is expected to provide a solid theoretical foundation for optimizing transmission characteristics and better meet user needs in practical applications. The rest of this paper is organized as follows: in section "Force analysis of cycloid reducer considering multi-source errors," the design of the cycloid reducer and the MSEs modeling of components are introduced. The non-linear force analysis methods of the cycloid pin gear mechanism, the output mechanism pin, and each bearing under the influence of MSEs are presented.

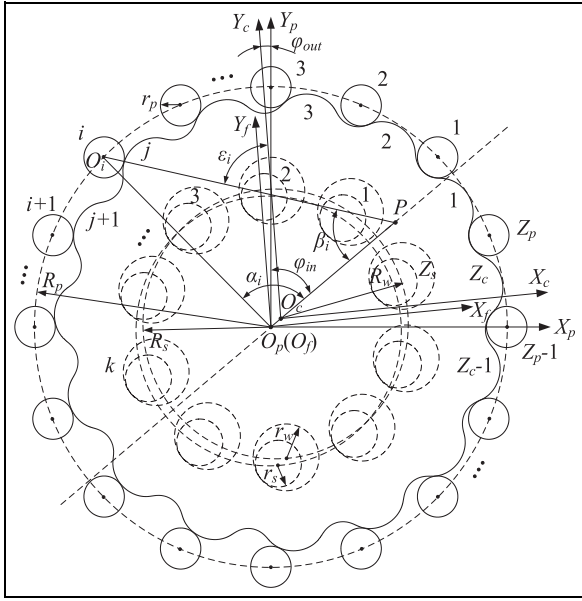


Figure 1. Cycloid pin gear transmission.

In section “Improved TEP method for the cycloid reducer,” the effects of MSEs, load-dependent friction power consumption, and load-independent friction power consumption are comprehensively considered, and an improved TEP method for predicting the transmission efficiency of the cycloid reducer under different operation conditions is proposed. In section “Transmission efficiency nonlinear characteristic prediction of the prototype,” the transmission efficiency of a reducer prototype is predicted based on the TEP method, and the effects of MSEs and load-independent power loss on its nonlinear characteristics are summarized. In section “Transmission efficiency nonlinear characteristic test of prototype,” the accuracy of the proposed TEP prediction is verified by prototype tests.

Force analysis of cycloid reducer considering multi-source errors

Multi-source errors modeling of the cycloid reducer

The standard tooth profile equation³⁷ is equation (1), where Z_c is the number of cycloid gear teeth, Z_p is the number of pin gear teeth, $i^H = Z_p/Z_c$, r_p is the pin radius, R_p is the pin distribution circle radius, K_1 is the short-amplitude coefficient, $K_1 = eZ_p/R_p$, and e is the crankshaft eccentricity. Define $S_c = 1 + K_1^2 - 2K_1 \cos \varphi_c$, and φ_c is the angular parameter.

In Figure 1, P is the node of gear meshing. The angle α_i between $O_p O_c$ and $O_p O_i$ is expressed as equation (2). In $\Delta O_i P O_p$, equation (3) can be deduced from the sine theorem. Equation (4) defines ε_i , the angle between the Y_c axis and the normal direction of the i -th pin's meshing point. Let the number of the nearest cycloid gear tooth and pin in the counterclockwise direction of the rotating arm $O_p O_c$ be 1, and the

number increases counterclockwise. j is the number of cycloid gear teeth in contact with the i -th pin.

$$\begin{cases} x_c = -(R_p - r_p S_c^{-\frac{1}{2}}) \sin[(1 - i^H)\varphi_c] \\ \quad - (e - K_1 r_p S_c^{-\frac{1}{2}}) \sin(i^H \varphi_c) \\ y_c = (R_p - r_p S_c^{-\frac{1}{2}}) \cos[(1 - i^H)\varphi_c] \\ \quad - (e - K_1 r_p S_c^{-\frac{1}{2}}) \cos(i^H \varphi_c) \end{cases} \quad (1)$$

$$\alpha_i = \text{mod} \left(\varphi_{in}, \frac{2\pi}{Z_p} \right) + \frac{2\pi(i-1)}{Z_p} \quad (2)$$

$$\sin \beta_i = \frac{\sin \alpha_i}{\sqrt{1 - 2K_1 \cos \alpha_i + K_1^2}} \quad (3)$$

$$\varepsilon_i = \pi - \beta_i - \varphi_{in} - \varphi_{out} \quad (4)$$

The designed cycloid reducer is exploded in Figure 2 and the design parameters are shown in Table 1. The cycloid gear and the input crankshaft are supported by bearings. The pins are inserted into the tooth grooves of the case. An output pin shaft is mounted within the output flange, with a pin shaft sleeve encasing the pin shaft.

Unavoidable MSEs require the cycloid gear tooth profile to be modified to meet the tooth-side clearance for assembly conditions. Equation (5) shows the commonly used isometric and offset combination modification,^{38–40} where ΔR_p is the isometric modification parameter, and Δr_p is the offset modification parameter. Define $S_{cm} = 1 + K_1' 2 - 2K_1' \cos \varphi_c$, and $K_1' = eZ_p/(R_p + \Delta R_p)$.

$$\begin{cases} x_{cd} = -[R_p + \Delta R_p - (r_p + \Delta r_p) S_{cm}^{-\frac{1}{2}}] \sin[(1 - i^H)\varphi_c] \\ \quad - (e - K_1' r_p S_{cm}^{-\frac{1}{2}}) \sin(i^H \varphi_c) \\ y_{cd} = [R_p + \Delta R_p - (r_p + \Delta r_p) S_{cm}^{-\frac{1}{2}}] \cos[(1 - i^H)\varphi_c] \\ \quad - (e - K_1' r_p S_{cm}^{-\frac{1}{2}}) \cos(i^H \varphi_c) \end{cases} \quad (5)$$

Cycloid reducer transmission performance depends on the pin gear, cycloid gear, and output pin shaft, and Figure 3 shows 13 kinds of MSEs associated with these components.⁴¹ Figure 4 shows the geometric characteristics of the MSEs of the cycloid drive mechanism. The ideal position O_{pi} of the i -th pin in Figure 4(a) is altered by pin distribution radius error δ_{Rpi} and pin tooth indexing error λ_{pi} , resulting in its revised position, O_{pti} . Equation (6) gives the pin center coordinates (x_{Oei} , y_{Oei}) in the fixed pin gear coordinate system $X_p O_p Y_p$. The pin radius error δ_{rpi} affects the pin's dimensions. The ideal cycloid gear rotation center position is O_c , while O_{ct} denotes the

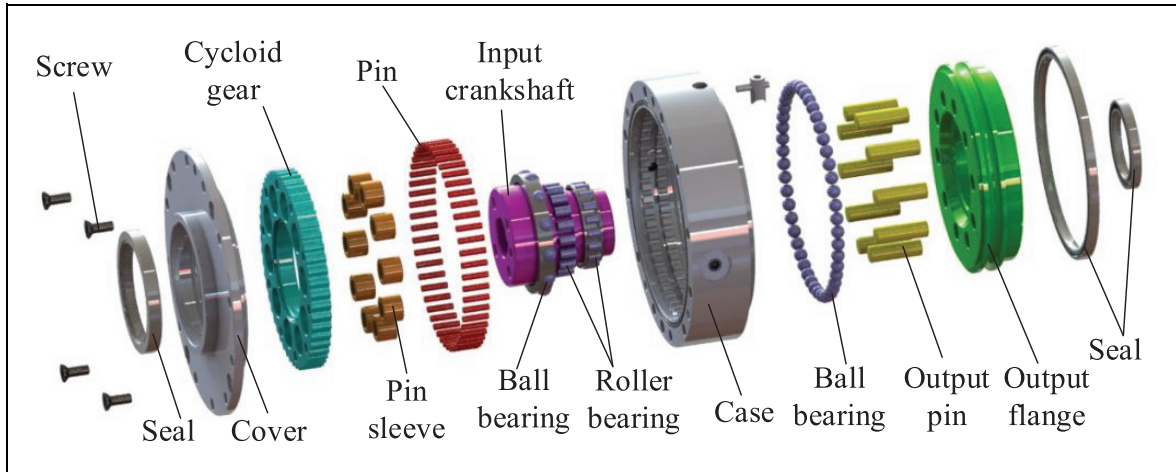


Figure 2. Exploded view of the cycloid reducer.

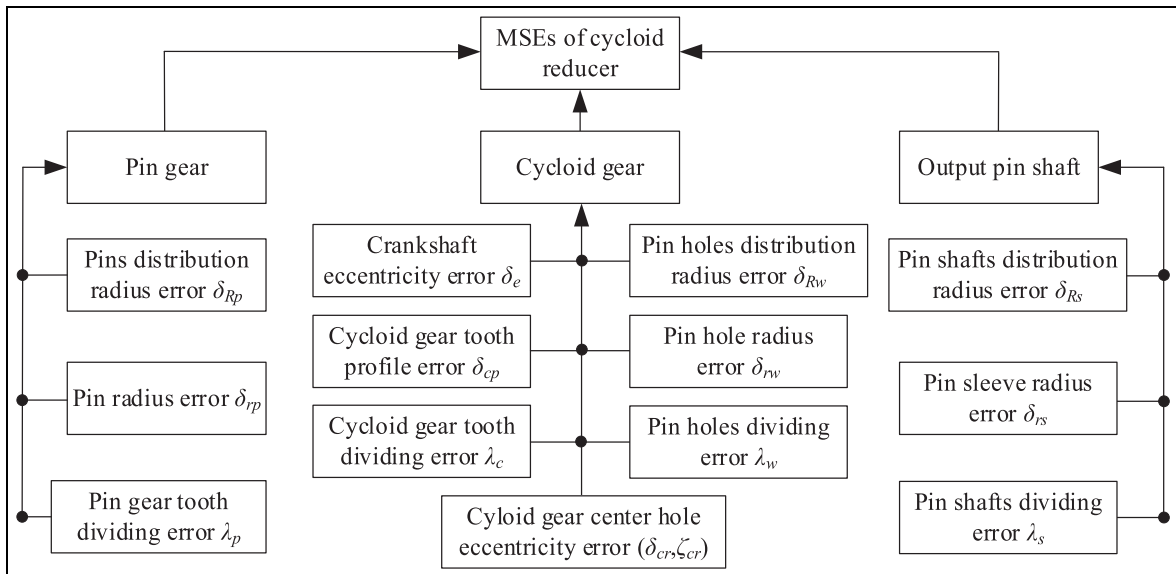


Figure 3. Multi-source errors of main components.

Table 1. Design parameters of the cycloid reducer.

Design parameters	Values
Cycloid gear tooth number Z_c	51
Pin gear tooth number Z_p	52
Pins distribution radius R_p	30 mm
Pin radius r_p	0.975 mm
Crankshaft eccentricity e	0.45 mm
Output pin shaft number Z_s	10
Pin hole radius R_w	4.5 mm
Pin sleeve radius R_s	4.05 mm

position affected by the crankshaft eccentricity error δ_e . The coordinates of O_{ct} under $X_p O_p Y_p$ can be calculated by equation (7). The j -th cycloid gear tooth profile is affected by the cycloid gear tooth indexing

error λ_{cj} . Figure 4(b) shows the eccentricity errors of the cycloid gear center hole (δ_{cr} , ζ_{cr}). Equation (8) expresses the coordinates (x_{cmj}, y_{cmj}) of j -th cycloid gear tooth profile in $X_p O_p Y_p$. Here, δ_{cr} is the center hole eccentricity error, ζ_{cr} is the phase angle parameter of the eccentricity, and (x_{cmj}, y_{cmj}) are the manufactured tooth profile coordinates in the $X_c O_c Y_c$.

$$\begin{cases} x_{Oei} = (R_p + \delta_{Rpi}) \sin \left(\frac{2\pi(1-i)}{Z_p} + \lambda_{pi} \right) \\ y_{Oei} = (R_p + \delta_{Rpi}) \cos \left(\frac{2\pi(1-i)}{Z_p} + \lambda_{pi} \right) \end{cases} \quad (6)$$

$$\begin{cases} x_{Oci} = (e + \delta_e) \sin \varphi_{in} \\ y_{Oci} = (e + \delta_e) \cos \varphi_{in} \end{cases} \quad (7)$$

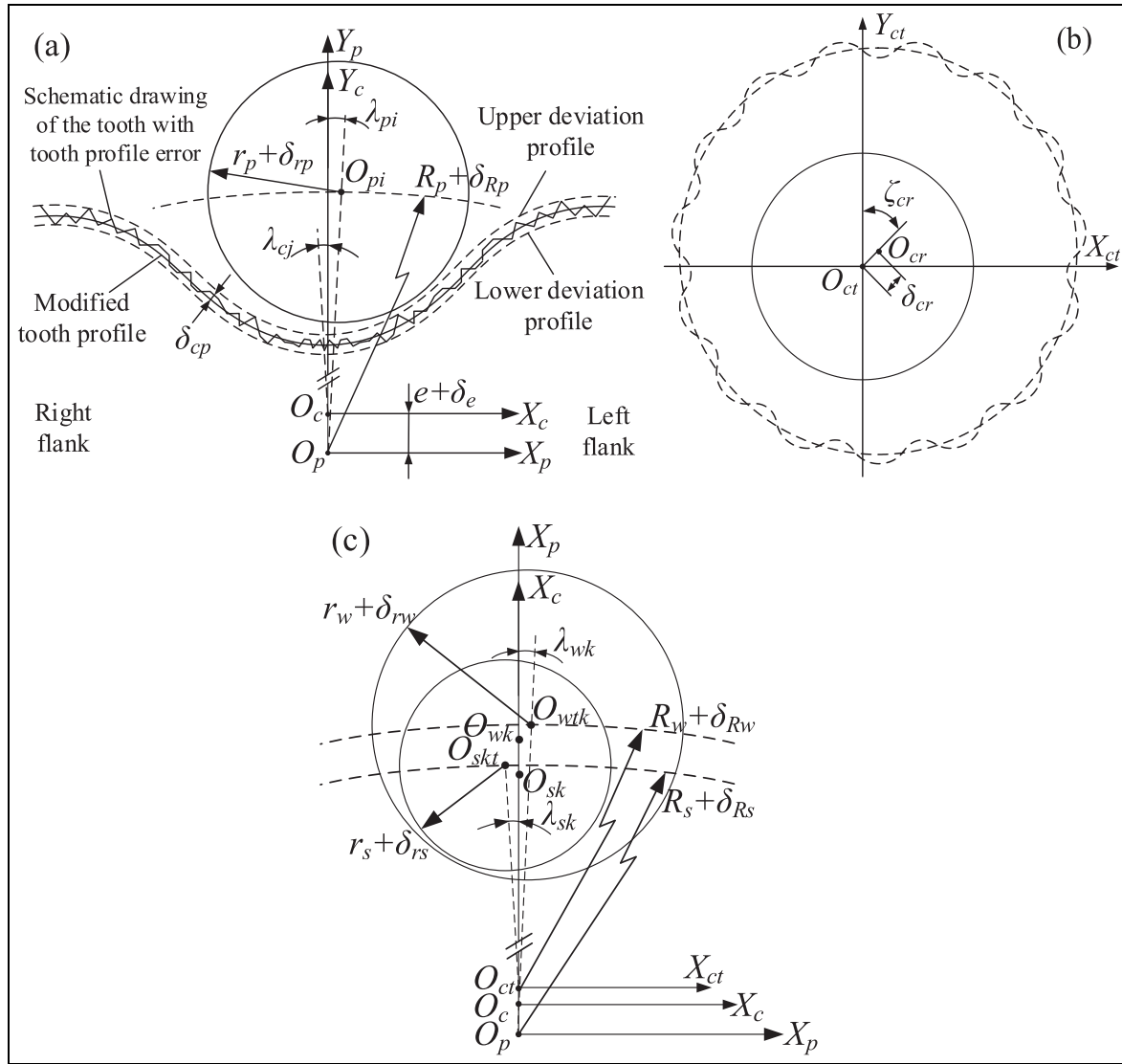


Figure 4. Geometric characteristics of: (a) MSEs of j -th cycloid gear tooth and i -th pin gear, (b) center hole eccentricity error, and (c) MSEs of the output mechanism.

$$\begin{cases} x_{cmpj} = (x_{cmj} + \delta_{cr} \sin \xi_{cr}) \cos(\varphi_{out} + \lambda_{cj}) \\ \quad - (y_{cmj} + \delta_{cr} \cos \xi_{cr}) \sin(\varphi_{out} + \lambda_{cj}) + x_{O_{ct}} \\ y_{cmpj} = (x_{cmj} + \delta_{cr} \sin \xi_{cr}) \sin(\varphi_{out} + \lambda_{cj}) \\ \quad + (y_{cmj} + \delta_{cr} \cos \xi_{cr}) \cos(\varphi_{out} + \lambda_{cj}) + y_{O_{ct}} \end{cases} \quad (8)$$

$$\delta_{cpN} = A_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega\varphi_c) + b_n \sin(n\omega\varphi_c)] \quad (9)$$

As shown in Figure 4(a), the schematic drawing of the tooth with tooth profile error is usually the curve between the upper and the lower equidistant deviation curves of the designed tooth profile.⁴² Let δ_{cpN} denote the deviation between the manufactured tooth profile and the modified tooth profile at the N -th meshing point in the normal direction ($N = 1, 2, \dots$,

N_c). The tooth profile error δ_{cpN} is influenced by the grinding gear dressing error,⁴³ making forward modeling difficult during design. In single-tooth cycloid gear profile grinding, each tooth's profile error is believed to be the same. Thus, the equivalent modeling of δ_{cpN} can be achieved by Fourier fitting of the measuring results of the manufactured cycloid gear, which is expressed as equation (9). n is the harmonic order, a_n and b_n are the amplitudes of harmonic orders, and φ_c is the tooth profile design parameter. The normal direction ε_i of the meshing point and the design parameter φ_{cj} of the j -th cycloid gear profile are related as $\varepsilon(\varphi_{cj})$. To account for indexing error λ_{cj} and tooth profile error δ_{cpN} , the j -th cycloid tooth profile (x_{cmj}, y_{cmj}) in $X_c O_c Y_c$ is given in equation (10). (x_{cdj}, y_{cdj}) is the modified tooth profile.

Figure 4(c) illustrates the geometric characteristics of the MSEs in the output mechanism. Using the subscript $k = 1, 2, \dots, Z_s$, the value of the same error

$$R_m = \begin{bmatrix} \cos \gamma_m & -\sin \gamma_m & 0 & (1 - \cos \gamma_m)(e + \delta_e) \sin \varphi_{in} + (e + \delta_e) \cos \varphi_{in} \sin \gamma_m \\ \sin \gamma_m & \cos \gamma_m & 0 & (1 - \cos \gamma_m)(e + \delta_e) \cos \varphi_{in} - (e + \delta_e) \sin \varphi_{in} \sin \gamma_m \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (15)$$

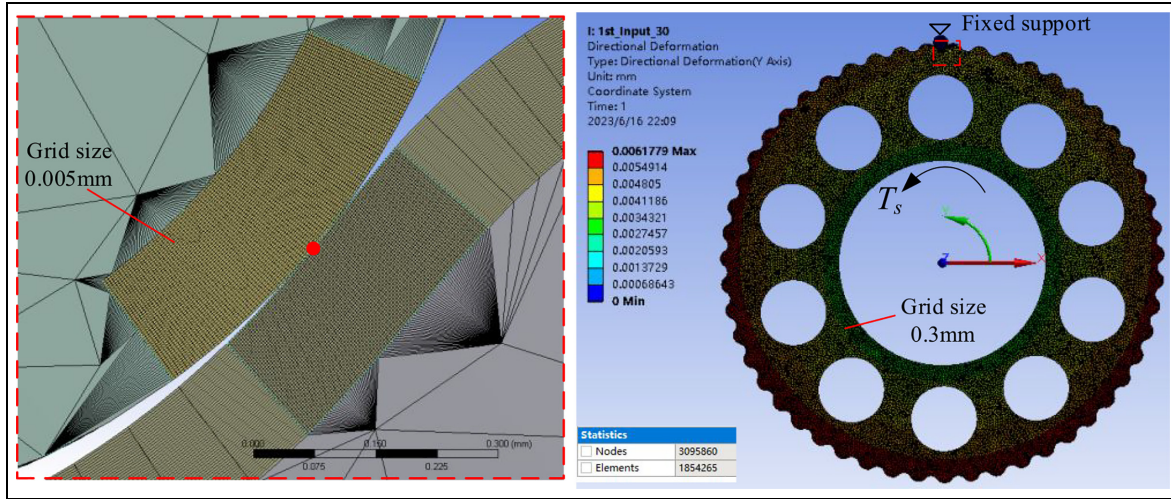


Figure 6. FEM analysis of the cycloid gear contact deformation.

$$r_{pcm}(\varphi_{in}) = [x_{cmptj}, y_{cmptj}, 0, 1]^T = R_m r_c(\varphi_{in}) \quad (16)$$

$$\gamma_{ci} = \min(\overline{G_{ti}M_{tiN}}/\overline{O_{ct}M_{tiN}}), (N = 1, 2, \dots, N_c) \quad (17)$$

$$\begin{cases} \overline{G_{tiN}M_{tiN}} = \overline{H_pM_{tiN}} - \overline{H_pG_{tiN}} \\ \overline{H_pM_{tiN}} = \sqrt{\overline{O_{pti}M_{tiN}^2} - \overline{O_{pti}H_p^2}} \\ \overline{H_pG_{tiN}} = \sqrt{(r_p + \delta_{rp})^2 - \overline{O_{pti}H_p^2}} \end{cases} \quad (18)$$

$$l_{ci} = \frac{eZ_c \sin \alpha_i}{\sqrt{1 - 2K_1 \cos \alpha_i + K_1^2}} \quad (19)$$

$$l_{cfi} = \sqrt{\overline{O_{ct}O_{pti}^2} - l_{ci}^2 - r_p^2} \quad (20)$$

Secondly, the torsional stiffness of the cycloid pin gear meshing needs to be modeled. It is assumed that the torsional stiffnesses of different numbers of cycloid gear teeth and pins are approximately the same. Therefore, based on the finite element method (FEM), the equivalent modeling of the torsional stiffness for the single-tooth meshing of the cycloid pin gear can be achieved by selecting the corresponding meshing states sequentially at an interval of 10° of input shaft rotation angle. As shown in Figure 6, the input rotation angle $\varphi_{in} = 30^\circ$ is taken as an example.

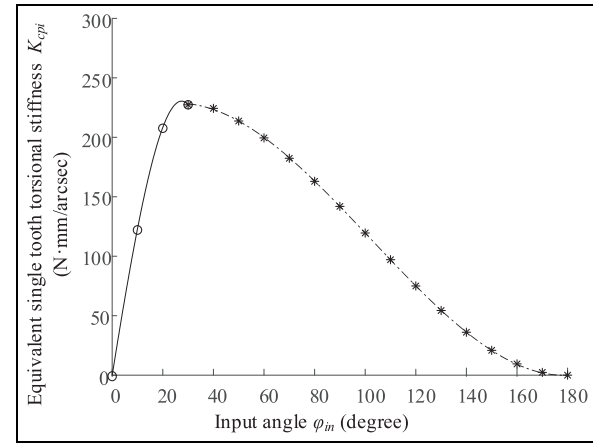


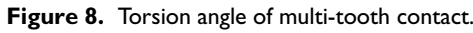
Figure 7. Equivalent torsional stiffness K_{cpi} .

2.06×10^6 Mpa, and Poisson's ratio is 0.3. Fixed constraints are added to the pin, and displacement constraints are added to the cycloid gear so it can only revolve around the central hole. The contact behavior between the cycloid teeth and the pin is set to frictionless. Add torque to the cycloid gear, analyze the torsional deformation of the cycloid gear, and then calculate the torsional stiffness. Figure 7 shows the modeling results of the cycloid pin gear single tooth contact torsional stiffness by changing the input angle and using the same setup approach. Equation (21) is the result of the segmented polynomial least squares fitting of K_{cpi} to φ_{in} .

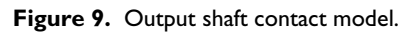
$$K_{cpi} = \begin{cases} -0.004697\varphi_{in}^3 - 0.04869\varphi_{in}^2 + 13.27\varphi_{in}, & 0 \leq \varphi_{in} \leq 30 \\ 0.0001304\varphi_{in}^3 - 0.04003\varphi_{in}^2 + 1.839\varphi_{in} + 205.55, & 30 < \varphi_{in} \leq 180 \end{cases} \quad (21)$$

The local contact area between the cycloid gear and the pin is set to be 0.005mm hexahedral mesh, and the rest is set to be 0.3mm tetrahedral mesh. The material modulus of elasticity for each component is

Ultimately, an analytical LTCA algorithm can be proposed. The algorithm needs to sort the γ_{ci} ($i = 1, 2, \dots, Z_p$) obtained by equation (17) from smallest to largest values and get a new array γ_{Tcn} ($n = 1, 2, \dots$,



Both the pin-type output mechanism and the cycloid drive mechanism are multi-body contact models under load conditions. In Figure 9, the input angle is φ_{in} , $X_p O_p Y_p$ is the fixed coordinate system of the pin gear, $X_f O_f Y_f$ is the rotational coordinate system of the output flange, O_{stk} is the actual center point of the k th pin with error, O_{wtk} is the actual center point of the k -th pin hole with error, and there initial clearance between each pin shaft sleeve and the pin hole.



Next, model the torsional stiffness of a single pin shaft in contact with the pin hole. Local contact deformation u_{Lsk} and global structural deformation u_{Gsk} make up the contact deformation u_{sk} . u_{Lsk} can be

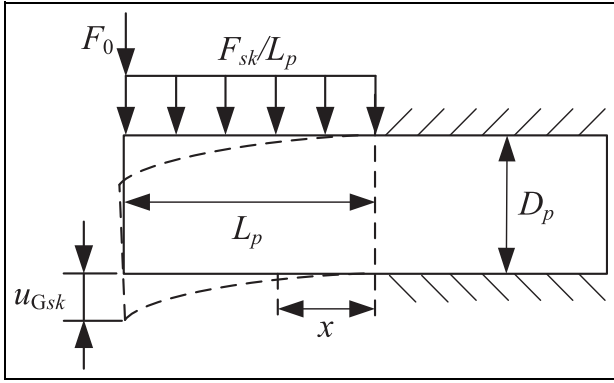


Figure 10. Structural deformation of the pin shaft.

calculated by equation (29).⁴⁴ The pin shaft structure belongs to the short cantilever beam. In Figure 10, the structural deformation u_{Gsk} needs to comprehensively consider the influence of the bending moment acting on the pin shaft and the shearing force of the section. According to Castigliano's theorem,⁴⁵ by adding a load F_0 to the end of the cantilever, the deformation u_{Gsk} at the end of the cantilever beam can be expressed by the left part of equation (30). In equation (30), $M(x)$ is the section's bending moment along the length L_p , and $M(x) = -F_0x - 0.5F_{sk}x^2/L_p$. $F(x)$ is the shearing force, and $F(x) = F_0 - F_{sk}x/L_p$. I is its moment of inertia, and A is its area. The calculation requires first taking the partial derivatives of $M(x)$ as well as c to obtain $\partial M(x)/\partial F_0 = -x$ and $\partial M(x)/\partial F_0 = -1$. At this point, for ease of calculation, let $F_0 = 0$. Then $M(x) = -0.5F_{sk}x^2/L_p$ and $F(x) = -F_{sk}x/L_p$. Bringing the calculation results from the above process into equation (30), the deformation u_{Gsk} can be expressed as the right part of equation (30). Equation (31) expresses the torsional stiffness of the k -th pin shaft.

$$u_{Lsk} = \frac{2F_{sk}(1-\nu^2)}{\pi L_p E} \ln \frac{6.59L_p^3 E(r_w - r_s)}{2(1-\nu^2)F_{sk}r_s r_w} \quad (29)$$

$$u_{Gsk} = \int_0^{L_p} \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F_0} dx + \int_0^{L_p} \frac{F(x)}{GA} \frac{\partial F(x)}{\partial F_0} dx \Big|_{F_0=0} = \frac{F_{sk}L_p^3}{8EI} \left(1 + \frac{24EI}{5GAL_p^2} \right) \quad (30)$$

$$K_{Tsk} = \frac{20\pi F_{sk}l_{sk}^2}{u_{Lsk} + u_{Gsk}} \quad (31)$$

Similarly, the contact force analysis of the output mechanism is achieved. Sort the value of γ_{sk} from small to large to obtain a new array γ_{Tsn} ($n = 1, 2, \dots, Z_s/2$). γ_{Tsn} denotes the minimum angle of rotation required for the simultaneous contact of n pin shafts

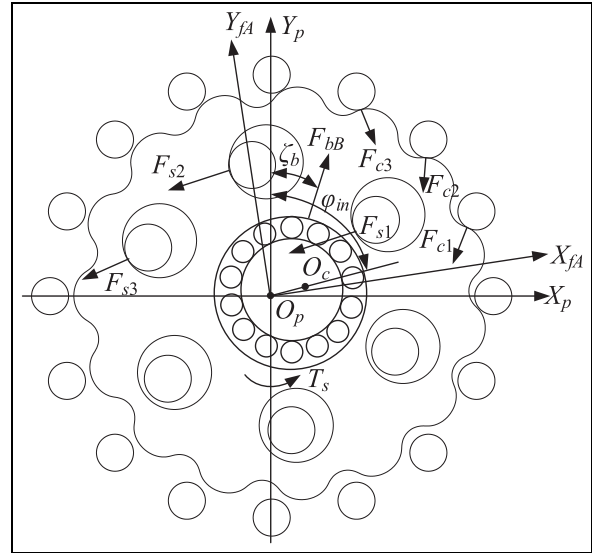


Figure 11. Cycloid gear and crankshaft bearing force.

and pin holes. $\Delta\gamma_{Tsq}$ indicates the rotational angle of the output flange when the number of pin shafts involved in the load increases from q to $q + 1$. Equation (32) relates the number n to the required minimum torque T_{sn} . The torsion angle γ_{TRs} of the output mechanism under load T_R can be calculated by equation (33). In equation (33), let the value of T_{sn} that is smaller than T_R and closest to T_R be T_{sMs} . M_s is the number of pin shafts that participate in load transmission. The contact force F_{sk} of the k -th pin shaft is calculated by equation (34).

$$T_{sn} = \begin{cases} 0, & n = 1 \\ \sum_{q=1}^{n-1} \left(\Delta\gamma_{Tsq} \left(\sum_{q=1}^{n-1} K_{Tsq} \right) \right), & n > 1 \end{cases} \quad (32)$$

$$\gamma_{TRs} = \gamma_{TcM_s} + \frac{T_R - T_{sM_s}}{\sum_{q=1}^{M_s-1} K_{Tsq}} \quad (33)$$

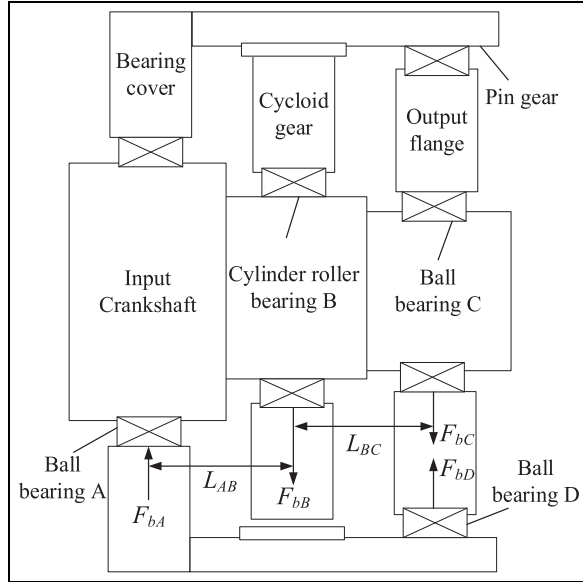
$$F_{sk} = \frac{K_{Tsk}(\gamma_{TRs} - \gamma_{sk})}{l_{sk}} \quad (34)$$

Force analysis of the bearings

Figure 11 shows a schematic diagram of the forces on the cycloid gear and bearing when the output load torque is T_s . Let the radial force of the crankshaft bearing be F_b , and the angle between F_b and Y_p axis be ζ_b . The combined force of F_{sk} and F_{ci} influences F_b and ζ_b , which are decomposed into F_{bx} and F_{by} in the $X_pO_pY_p$ coordinate system. Equations (35) and (36) are satisfied when the internal static force of the reducer is balanced. In equation (35), ε_{pk} is the angle

Table 2. Load-dependent power loss parameters of the cycloid reducer.

Power loss parameter	P_c	P_s	P_{bA}
Function	$\sum_{q=1}^{M_c} \mu_{ci} F_{ci} l_{cfi} \omega_{out}$	$\sum_{q=1}^{M_s} \mu_{sk} F_{sk} l_{sfk} \omega_{out}$	$\mu_{bA} F_{bA} D_{bA} \omega_{in}$
Power loss parameter	P_{bB}	P_{bC}	P_{bD}
Function	$\mu_{bB} F_{bB} D_{bB} (\omega_{in} + \omega_{out})$	$\mu_{bC} F_{bC} D_{bC} (\omega_{in} + \omega_{out})$	$\mu_{bD} F_{bD} D_{bD} \omega_{out}$

**Figure 12.** Force analysis of each bearing.

between F_{sk} and Y_p axis, and ε_{pi} is the angle between F_{ci} and Y_p axis.

$$\begin{cases} F_{bx} + \sum F_{ci} \sin \varepsilon_{pi} + \sum F_{sk} \sin \varepsilon_{pk} = 0 \\ F_{by} + \sum F_{ci} \cos \varepsilon_{pi} + \sum F_{sk} \cos \varepsilon_{pk} = 0 \end{cases} \quad (35)$$

$$\zeta_b = \text{atan2}(F_{bx}, F_{by}) \quad (36)$$

In Figure 12, each bearing inside the reducer is defined as bearing ABCD. The axial distance between bearing A and bearing B is L_{AB} , whereas the axial distance between bearing B and bearing C is L_{BC} . Bearings C and D are approximated to be in the same axial position inside the reducer, that is, the axial distance between bearings B and D is also L_{BC} . The parameters for calculating the frictional torque of the bearing are distinguished by subscripts. Take bearing B as an example, where bearing B is subjected to a radial force F_{bB} , and the diameter of the outer ring of bearing B is D_{bB} . For the force analysis of each bearing inside the reducer, the radial force of bearing B between the cycloid gear and the crankshaft is calculated according to the load balancing condition, and then the radial force in the remaining bearings is calculated from the force balance condition of the

crankshaft. The radial force of each bearing is expressed by equation (37).

$$\begin{cases} F_{bB} = \sqrt{F_{bx}^2 + F_{by}^2} \\ F_{bA} = F_{bB} L_{AB} / (L_{AB} + L_{BC}) \\ F_{bC} = F_{bB} L_{BC} / (L_{AB} + L_{BC}) \\ F_{bD} = F_{bB} L_{BC} / (L_{AB} + L_{BC}) \end{cases} \quad (37)$$

Improved TEP method for the cycloid reducer

Modeling of power losses

The power loss of the gear transmission system can be divided into load-dependent and load-independent power loss.⁴⁶ Load-dependent power loss is caused by the contact between the transmission mechanism meshing when the reducer transmits torque, also known as frictional power loss, which is an essential part of the power loss of the reducer. The load-dependent power loss of the designed cycloid reducer comes from the friction power loss P_c of the cycloid drive mechanism, the friction power loss P_s of the output mechanism, and the friction power loss P_{b*} of each bearing (* denotes the subscripts of the bearings' classification ABCD). Therefore, the reducer load-dependent frictional power loss P_L can be expressed by equation (38).

$$P_L = P_c + P_s + P_{bA} + P_{bB} + P_{bC} + P_{bD} \quad (38)$$

The equation for each power loss parameter in equation (38) are shown in Table 2. It can be seen that there is a linear relationship between the value of the load-dependent power loss generated by each component and the input speed, as well as the coefficient of friction. The transmission ratio of the reducer is 51, and the angular velocity of the cycloid gear rotation is reduced after deceleration, so the change in the coefficient of friction on the power loss generated by the cycloid drive mechanism produces a very low impact on the entire reducer. For the meshing friction coefficient of the cycloid drive mechanism, the value of $\mu_{ci} = 0.05$ is taken, and similarly, for the pin output mechanism, which also belongs to the low-speed transmission parts, the contact friction coefficient is taken as $\mu_{sk} = 0.05$. According to the guidance of

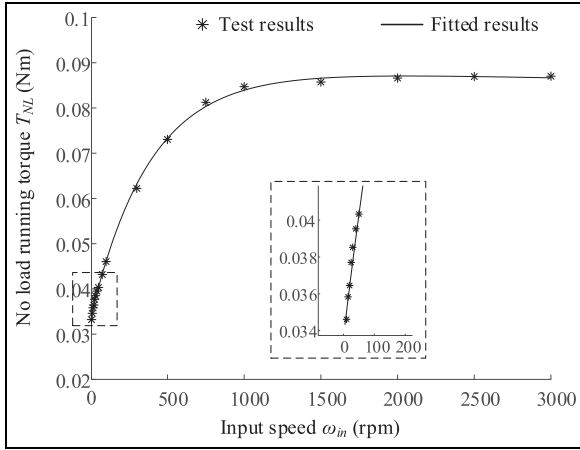


Figure 13. Load-independent frictional torque test results.

SKF and JTEKT, the friction coefficient of cylindrical roller bearings is selected as $\mu_{bB} = 0.001$, and the friction coefficients of the remaining ball bearings are $\mu_{bA} = \mu_{bC} = \mu_{bD} = 0.0008$.

$$T_{NL} = 0.089 \exp(-7.84 \times 10^{-6} \omega_{in}) - 0.055 \exp(-0.026 \omega_{in}) \quad (39)$$

$$P_{NL} = T_{NL} \omega_{in} \quad (40)$$

The load-independent loss of the cycloid reducer is the loss caused by the rotation of the system when the torque is not transmitted. The load-independent power loss of the reducer comes from several sources, including gear loss, bearing loss, sealing loss, and auxiliary components loss. The cycloid reducer transmission efficiency model for load-independent power loss has been studied less. Some research studies^{30,31} have shown that under different speed conditions without load, the load-independent friction torque of the reducer shows a specific non-linear relationship with the speed. Since the forward mathematical modeling

of load-independent friction torque is very complicated, this study will be based on testing the prototype for its equivalent modeling. The test scheme will be introduced in section “Test results and discussions.” As shown in Figure 13, the test results of the internal load-independent friction torque of the reducer prototype under different input speeds without load are shown in Figure 13, and the relationship between the friction torque and input speed can be equivalently modeled by equation (39) through the exponential fitting. The load-independent power loss of the reducer is calculated by equation (40).

Improved TEP method

Based on the modeling results of load-dependent and load-independent power loss, an improved transmission efficiency prediction (TEP) method considering MSEs is proposed, and the specific process of the TEP method is shown in Figure 14.

- (1) Input the design parameters and MSEs parameters of the reducer, set the initial step of the input shaft angle $N = 0$, the input angle step is $\Delta\varphi_{in}$, the load at the output end is T_{out} , and the initial input shaft angle is $\varphi_{in} = 0$.
- (2) Calculate the meshing force F_{ci} of the cycloid drive mechanism from equation (24). Calculate the contact force F_{sk} of the output mechanism from equation (34). Calculate the radial force of each bearing from equation (37).
- (3) Calculate the load-dependent power loss from equation (38) and the load-independent power loss from equation (40). Calculate the transient transmission efficiency from equation (41).
- (4) Determine whether the output of the reducer has rotated one cycle, that is, satisfy the condition $\varphi_{in}/Z_c \geq 2\pi$, then terminate the TEP analysis. Otherwise, gradually increase the input angle and repeat steps (2) and (3) until the termination condition is satisfied.

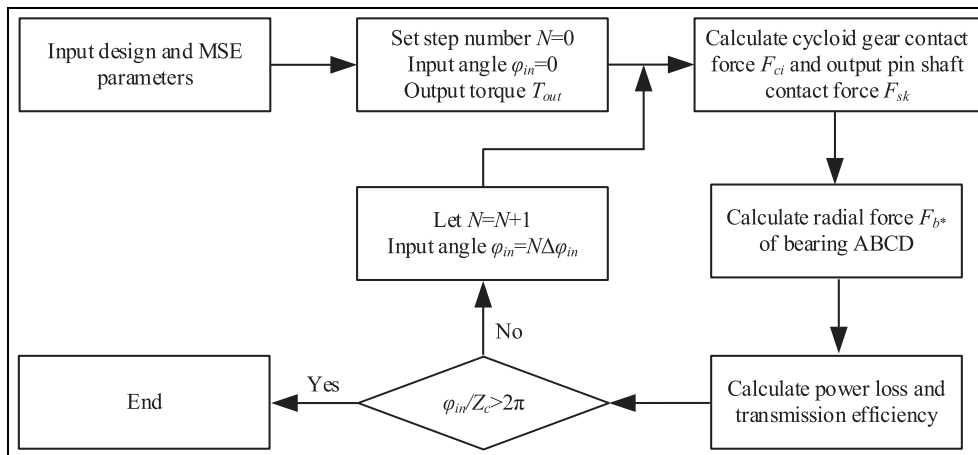


Figure 14. Flow of the TEP method.

$$\eta_{TEP} = \frac{T_{out}\omega_{out}}{T_{out}\omega_{out} + P_L + P_{NL}} \times 100\% \quad (41)$$

The $T_{out}\omega_{out}$ in equation (41) is the actual output power of the reducer, while the denominator part is the sum of the actual input power of the reducer. According to the above TEP model, calculated from equation (41) is the transient transmission efficiency of the entire reducer in the range of one revolution of the output flange. The maximum value of the transient transmission efficiency of the reducer is denoted as $\max(\eta_{TEP})$, the minimum value is denoted as $\min(\eta_{TEP})$, and the average value is denoted as $\bar{\eta}_{TEP}$. The difference between the maximum and minimum values of the transient transmission efficiency is denoted as $\Delta\eta_{TEP}$.

At present, the conventional analysis methods for the transmission efficiency of cycloid reducers do not adequately consider the impact of MSEs, and they fail to account for and effectively model load-independent power losses. In contrast, the TEP method proposed in this study considers the effects of MSEs, load-dependent power losses, and load-independent power losses, making it an improved prediction method. The TEP method facilitates further research on the nonlinear characteristics of the reducer's transmission efficiency under different operating conditions and theoretically enhances the accuracy of transmission efficiency predictions. To validate the superiority of this improved TEP method over conventional methods, in-depth studies are required in conjunction with performance prediction examples using a prototype. Specifically, when investigating the nonlinear transmission efficiency of the reducer under constant operating conditions, the transient traditional efficiency η_{TEP} can be used to adequately describe its performance. On the other hand, for studying the nonlinear transmission efficiency under varying operating conditions, the average value of the transmission efficiency $\bar{\eta}_{TEP}$ and the fluctuation range $\Delta\eta_{TEP}$ can be

Table 3. Tolerance design results.

Parameters	Tolerance
Pin distribution radius error δ_{Rp}	[5,10] μm
Cycloid gear tooth profile error δ_{cpN}	[−5,5] μm
Pin radius error δ_{rp}	[−1.5,0] μm
Cycloid gear tooth dividing angle error λ_c	[−15,15] arcsec
Pin gear tooth dividing angle error λ_p	[−15,15] arcsec
Crankshaft eccentricity error δ_e	[−4,4] μm
Cycloid gear center hole eccentricity error δ_{cr}	[0,2] μm
Output flange pin distribution radius error δ_{Rs}	[−2,2] μm
Cycloid gear pin hole distribution radius error δ_{Rw}	[−2,2] μm
Output pin sleeve radius error δ_{rs}	[−2,0] μm
Output pin hole radius error δ_{rw}	[3,5] μm
Output pin shaft dividing angle error λ_s	[−15,15] arcsec
Cycloid gear pin hole dividing angle error λ_w	[−15,15] arcsec

employed to examine the impact of changing conditions on the reducer's performance.

Transmission efficiency nonlinear characteristic prediction of the prototype

Transmission efficiency nonlinear characteristic due to MSEs

The values of the MSEs of components can be constrained by reasonable tolerance design. As shown in Table 3 regarding the ISO standard, the tolerance is designed in the range of IT5–IT6, and the modification parameters in equation (5) are designed as $\Delta R_p = -0.008 \text{ mm}$ and $\Delta r_p = -0.002 \text{ mm}$.

Each component is measured by precision measuring instruments. Take the measurement process of the cycloid gear error as an example. As shown in Figure 15(a), the MSEs of the cycloid gear are

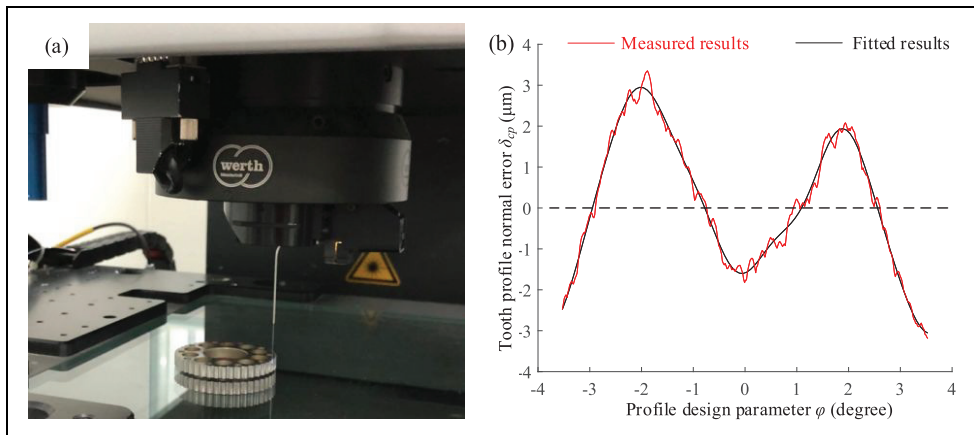


Figure 15. (a) MSEs measurement of cycloid gear and (b) measurement results of tooth profile error δ_{cpN} .

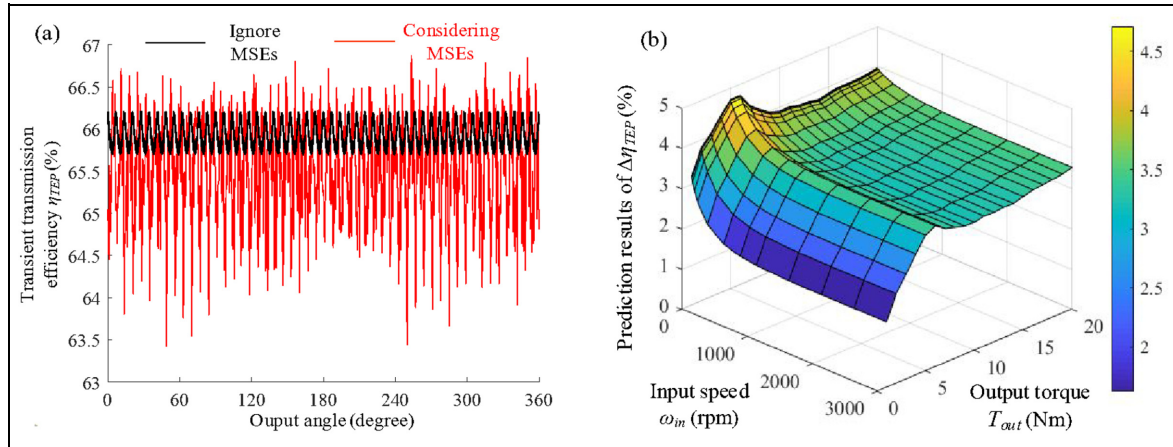


Figure 16. Considering the effect of multiple sources of error on the fluctuation of transmission efficiency of the reducer at: (a) rated operating conditions and (b) multiple operating conditions.

Table 4. Fitting parameters for tooth profile error.

N	1	2	3	4	5	6
a_n	0.42	-2.23	-0.33	0.05	-0.07	-0.10
b_n	-0.66	-0.29	-0.12	0.07	0.08	0.06

measured by the Werth VideoCheck-S model CMM. The tooth profile error of each cycloid gear tooth is approximately the same. The red curve in Figure 15(b) is the measured tooth profile error, and the black curve is the fitted result of δ_{cpN} shown in equation (42) and Table 4. The crankshaft eccentricity error δ_e is $2.5 \mu\text{m}$, the cycloid gear center hole eccentricity error δ_{ro} is $1.8 \mu\text{m}$, and the cycloid gear center hole eccentricity error phase angle ζ_{ro} is 30.6° . Cycloid pin bore distribution radius error δ_{Rw} is $0.5 \mu\text{m}$, pin bore radius error δ_{rw} is $4.4 \mu\text{m}$, and output flange pin distribution radius error δ_{Rs} is $1.8 \mu\text{m}$. The rest of the MSE measurement results are shown in Appendix Tables A1 to A3, respectively.

$$\delta_{cpN} = 0.014 + \sum_{n=1}^6 [a_n \cos(48.3n\varphi) + b_n \sin(48.3n\varphi)] \quad (42)$$

The rated operating conditions of the reducer are 3000 rpm and 20 Nm load, respectively. In order to verify the influence of MSEs on the nonlinearity of the reducer transmission efficiency, as shown by the red line in Figure 16(a), the design parameters of the reducer and the MSEs data obtained from measurements are substituted into the proposed TEP method and the transient transmission efficiency of the reducer prototype is predicted, in which the load-independent power loss P_{NL} is calculated from equation (40). The black lines in Figure 16(a) show the predicted transient transmission efficiency of the reducer

without considering the MSEs, that is, all the MSEs data substituted into the TEP method are set to 0. From the prediction results in Figure 16(a), the transmission efficiency of the prototype considering the MSEs is $\max(\eta_{TEP}) = 66.9\%$, $\min(\eta_{TEP}) = 63.4\%$, $\Delta\eta_{TEP} = 3.5\%$, and $\bar{\eta}_{TEP} = 65.5\%$. In contrast, the transmission efficiency of the prototype without considering MSEs is $\max(\eta_{TEP}) = 66.2\%$, $\min(\eta_{TEP}) = 65.7\%$, $\Delta\eta_{TEP} = 0.5\%$, $\bar{\eta}_{TEP} = 65.9\%$.

The transient transmission efficiency without considering MSEs varies periodically, but since the fluctuation range of the black line is small, it is feasible to approximate the efficiency as constant under unchanging operational conditions. This justifies the conventional method of using the average value to assess the reducer's transmission efficiency. However, at rated conditions, the prototype's transient efficiency fluctuation is seven times larger than that of the prototype without MSEs consideration, which highlights the necessity of accounting for the nonlinear transient transmission efficiency characteristics induced by MSEs in practical applications. Because the transient transmission efficiency caused by MSEs will directly affect the smooth operation of the mechanical transmission system. In some cases, due to the transient transmission efficiency being too low, power loss increases rapidly due to the abnormal rise in temperature,⁴⁷ which will affect the thermal stability⁴⁸ of the system leading to abnormal wear and tear reducing the service life of the cycloid reducer. To this end, the effect of MSEs on the nonlinear fluctuations of the transmission efficiency of the reducer under different operating conditions needs to be further investigated.

Based on the proposed TEP method, the transient transmission efficiency fluctuations $\Delta\eta_{TEP}$ of the reducer are predicted for a wide range of operating conditions in the speed range of [5,3000] rpm and load range of [1,20] Nm. The prediction results are shown in Figure 16(b), where the maximum value of $\Delta\eta_{TEP}$ occurs under the conditions of a low input speed of

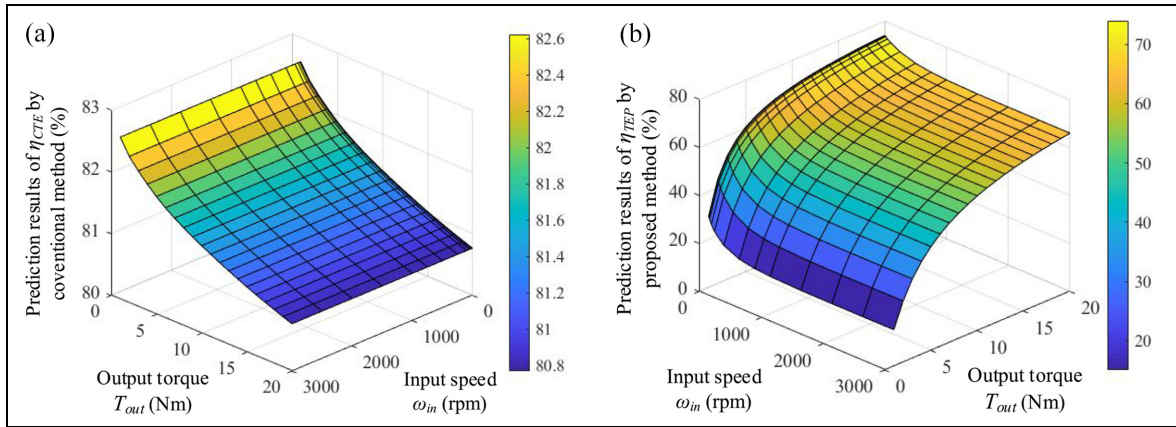


Figure 17. (a) When load-independent power losses are not taken into account and (b) reducer transmission efficiency when load-independent power losses are included $\bar{\eta}_{TEP}$.

5 rpm and a low load of 5 Nm, while the minimum value of $\Delta\eta_{TEP}$ occurs under the conditions of a high input speed of 3000 rpm and a low load of 1 Nm. When the load remains unchanged, the value of $\Delta\eta_{TEP}$ gradually decreases with the increase of input speed until it remains basically unchanged when the speed exceeds 500 rpm. When the input speed remains unchanged, the value of $\Delta\eta_{TEP}$ increases first and then decreases with the increase of load until it remains basically unchanged when the load exceeds 10 Nm.

Transmission efficiency nonlinear characteristic due to load-independent power losses

The conventional cycloid reducer transmission efficiency prediction method does not take into account the load-independent power loss. To explore the influence of load-independent power on the nonlinearity of the transmission efficiency of the reducer, it is necessary to predict the average value of the transmission efficiency of the reducer under different input speeds and output loads based on the conventional transmission efficiency prediction method and the proposed TEP method, respectively. Figure 17(a) shows the prediction results of the conventional method considering the effect of MSEs but does not include the load-independent power loss, that is, $P_{NL} = 0$. In Figure 17(a), the average transmission efficiency of the reducer under different operating conditions is more than 80% without considering the load-independent power loss. When the input speed remains unchanged, as the load increases, the average transmission efficiency of the reducer $\bar{\eta}_{TEP}$ decreases, but the reduction from the value of $\bar{\eta}_{TEP}$ is only close to 2%, which is negligible relative to the value of 80%. When the load remains unchanged, the value of $\bar{\eta}_{TEP}$ is constant with the change in input speed.

The prediction results of the proposed TEP method, which takes into account the effect of MSEs and load-independent power losses, are shown in

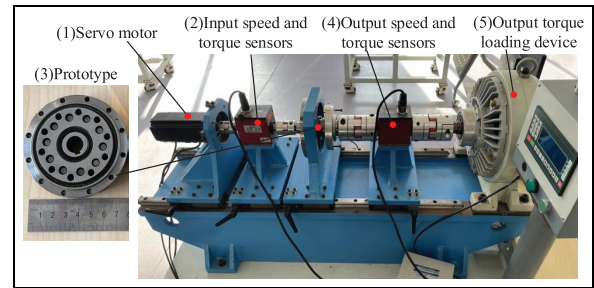


Figure 18. Transmission efficiency test rig and its components.

Figure 17(b). The predicted results show that the average transmission efficiency of the reducer does not exceed 75% for different speed and load conditions. When the input speed remains unchanged, the value of $\bar{\eta}_{TEP}$ increases as the load increases. When the load remains unchanged, the value of $\bar{\eta}_{TEP}$ decreases when the speed increases. In Figure 17(b), the average transmission efficiency $\bar{\eta}_{TEP}$ shows a significant nonlinear relationship with the load and speed conditions when load-independent power loss is considered.

Transmission efficiency nonlinear characteristic test of prototype

Test scheme

Prototype tests need to be carried out to verify the effect of MSEs and load-independent power loss on the nonlinear characteristics of the transmission efficiency of the prototype. The test rig for the transmission efficiency of the cycloid reducer prototype is shown in Figure 18, and its components includes: (1) Servo motor; (2) Input speed and torque sensors; (3) Prototype; (4) Output speed and torque sensors; (5) Output torque loading device. Therein, the prototype is mounted on the base of the test rig, and the two

sides of the input speed and torque sensor are connected to the servo motor and the input crankshaft of the prototype respectively through a coupling. The two sides of the output speed and torque sensor are connected to the ^{output} flange of the prototype and the output torque loading device. The torque measurement range of the input speed and torque sensor is 1 Nm, and the measurement accuracy is ± 0.001 Nm.⁴⁹ The torque measurement range of the output speed and torque sensor is 50 Nm, and the accuracy is ± 0.05 Nm.⁵⁰ Speed and load conditions during the tests are the same as those predicted by TEP method. The sampling frequency of the data will be appropriately adjusted according to the input speed of the servomotor in the range of [1,1000] Hz to ensure that the test results under different speed conditions include transient transmission efficiency information of at least 500 different output angular positions of the prototype. To maintain the accuracy of the test results, according to the test standard GB35089-2018, the fluctuation of the input and output torque caused by the coupling installation error is adjusted to be less than $\pm 1.5\%$ of the range of each sensor.

To validate the effectiveness of the proposed TEP method, the following three tests should be carried out:

- (1) Load-independent friction torque test at no load: Under the no output load condition of the prototype, when the prototype is running steadily at the input speed ω_{in} , the average value of the input torque $\bar{T}_{in}$ during a period is the load-independent friction torque of the prototype at the speed ω_{in} . By selecting different input speeds and testing the input torque of the prototype, the speed-torque curve shown in Figure 13 is obtained.
- (2) Transient transmission efficiency test under stable operating conditions: Add a constant value of input speed and load to the prototype and obtain the torque T_{in} and T_{out} of the input and output sensors during one circle rotation of the output flange under this stable working condition. Calculate the transient transmission efficiency η_{Test} of the prototype by equation (43). The difference between the maximum and minimum values of η_{Test} is estimated as $\Delta\eta_{Test}$.
- (3) Average transmission efficiency test under varying operating conditions: Different input speeds and loads conditions are selected, and the transient transmission efficiency η_{Test} is tested under each stable condition. The average value of η_{Test} is estimated as $\bar{\eta}_{Test}$, and the $\bar{\eta}_{Test}$ under different operating conditions are obtained.

$$\eta_{Test} = \frac{T_{out}}{Z_c T_{in}} \times 100\% \quad (43)$$

Test results and discussions

The transmission efficiency test results of the reducer prototype are shown in Figure 19(a) to (d). Figure 19(a) shows the test results of the transient transmission efficiency of the prototype at rated operating conditions, where $\max(\eta_{Test}) = 67.1\%$, $\min(\eta_{Test}) = 62.4\%$, $\bar{\eta}_{Test} = 65.0\%$, and the test results are highly consistent with the theoretical predictions of the proposed TEP method in Figure 16(a). Figure 19(b) shows the test results of the transient transmission efficiency fluctuation range $\Delta\eta_{Test}$ at different operating conditions, and the absolute value deviation of this result and the predicted result of Figure 16(b) is small, with a high degree of agreement, which verifies the validity of the proposed TEP method. At the same time, the test results show that when the load exceeds 50% of the rated load, the influence of MSEs on the transient transmission efficiency fluctuation of the reducer is more obvious. The transmission efficiency fluctuation range of the reducer under the same load and low input speed is generally higher than that under high input speed. Therefore, the influence of transient transmission efficiency fluctuation of the reducer in the case of a lower input speed and a larger load on the stability of the reducer needs to be paid attention to.

Figure 19(c) shows the test results of the average transmission efficiency $\bar{\eta}_{Test}$ of the prototype at different operating conditions. Figure 19(d) shows the percentage deviation of the predicted results $\bar{\eta}_{TEP}$ of the TEP method and the test results $\bar{\eta}_{Test}$, which are calculated by equation (44). The test results show that the accuracy of the proposed TEP method in predicting the nonlinear characteristics of the transmission efficiency of the cycloid reducer prototype under different input speeds and loads conditions is high when the effect of load-independent power loss is considered, and it has a better application value compared with the conventional method introduced in section "Transmission efficiency nonlinear characteristic due to load-independent power losses."

$$\delta_{TE} = (\bar{\eta}_{TEP} - \bar{\eta}_{Test}) / \bar{\eta}_{Test} \times 100\% \quad (44)$$

In summary, the proposed improved TEP method can effectively predict the transmission efficiency of the cycloid reducer. Based on the theoretical prediction and test results, the nonlinear characteristics of the transmission efficiency of the reducer caused by MSEs and load-independent power loss can be summarized.

Conclusion

In this study, an improved transmission efficiency prediction (TEP) method for the nonlinear characteristics of the cycloid reducer is proposed, and it reveals the influence of multi-source errors (MSEs) and

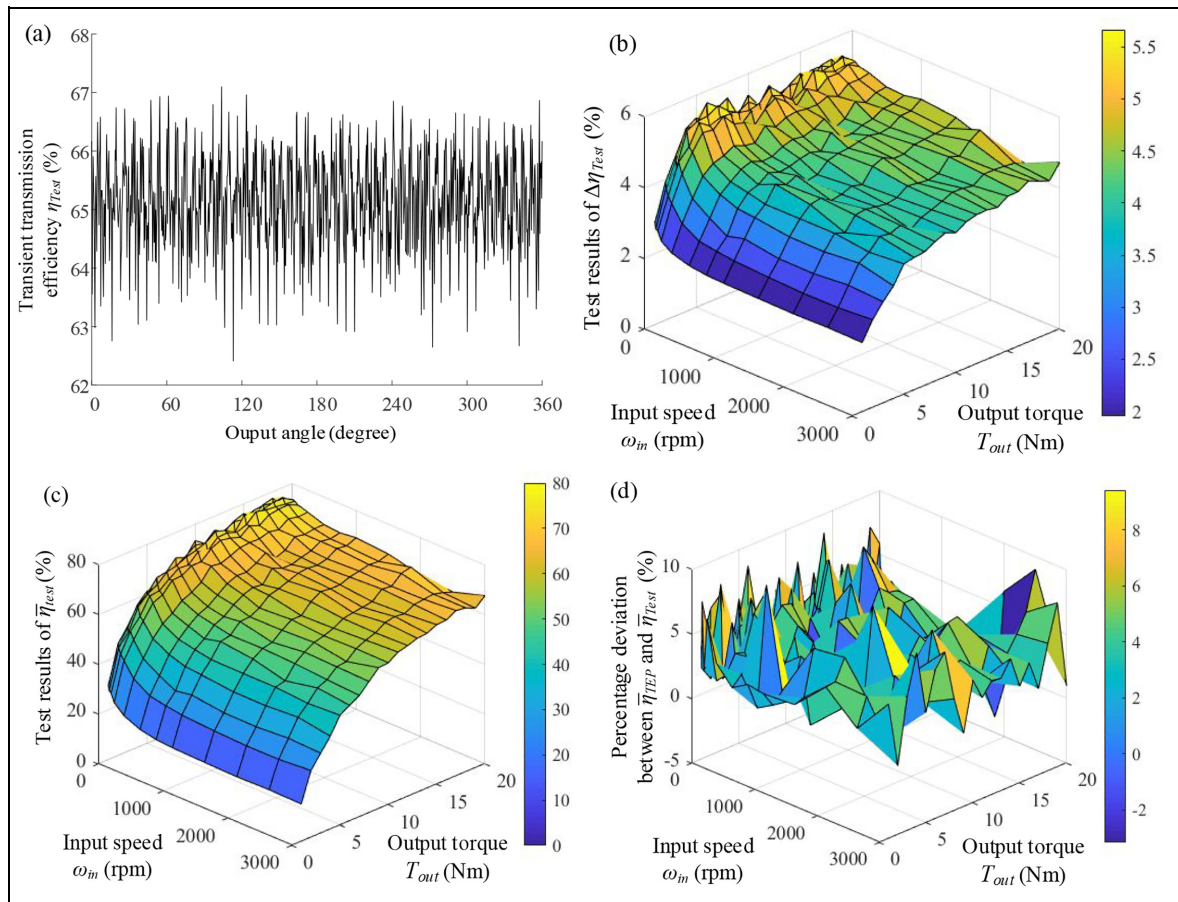


Figure 19. (a) Transient transmission efficiency test results at rated operating condition, (b) test results of fluctuation of transient transmission efficiency $\Delta\eta_{Test}$, (c) test results of the average transmission efficiency $\bar{\eta}_{Test}$, and (d) percentage deviation between the predicted and tested results.

load-independent power loss on the nonlinearity of the transmission efficiency of the reducer under different operating conditions. This study enriches the theoretical framework of accurate transmission efficiency prediction of cycloid reducers and ultimately enhances their optimization potential. The main results of this study are as follows:

- (1) The proposal of an analytical algorithm for force analysis considering MSEs: Based on the multi-body contact mechanism, an analytical algorithm is proposed for the force analysis of the cycloid drive mechanism and the output mechanism, respectively, taking into account the nonlinearities of the contact clearances caused by MSEs. Combined with the bearing force analysis, it achieves the effective analysis of the nonlinear force of the key components inside the reducer under different loading conditions. In addition, the MSEs of the components considered in this study are more reasonable, and the values of the MSEs are obtained by precision measurement, which helps to improve the accuracy of the prediction.
- (2) The proposal of an improved TEP method: Considering the influence of MSEs and load-

independent power loss, an improved TEP method of the reducer is proposed, which can provide a more accurate and comprehensive description of the transmission efficiency of the reducer under different operating conditions compared with the conventional method.

- (3) The investigation of the nonlinear characteristic of the transmission efficiency: Under constant operating conditions, the cycloid reducer exhibits nonlinear variations in transient transmission efficiency due to MSEs. This nonlinearity is pronounced at lower input speeds, highlighting the significance of cycloid reducer applications in such scenarios. Furthermore, due to the nonlinear relationship between load-independent power losses and input speed, the average transmission efficiency of the reducer gradually decreases with the increase of input speed under constant load. Conversely, it gradually increases with the increase of load under constant input speed.

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Appendix

Table A1. Components measurement results of δ_{rpi} and δ_{Rpi} .

<i>i</i>	1	2	3	4	5	6	7	8	9	10	11
δ_{rpi} (μm)	−0.7	−0.8	−1	−0.8	−1	−1.1	−0.6	−0.4	−0.9	−1.2	−0.3
δ_{Rpi} (μm)	9.4	7.2	8.8	5.6	7.6	7.9	7.6	6.6	6.9	9.1	8.3
<i>i</i>	12	13	14	15	16	17	18	19	20	21	22
δ_{rpi} (μm)	−1	−0.9	−0.9	−0.8	−1	−0.8	−0.7	−0.9	−0.5	−0.7	−0.9
δ_{Rpi} (μm)	8.0	7.4	7.8	6.9	6.7	6.9	5.5	8.7	9.6	7.7	8.4
<i>i</i>	23	24	25	26	27	28	29	30	31	32	33
δ_{rpi} (μm)	−0.9	−1.0	−0.8	−0.9	−1.0	−0.9	−0.7	−0.9	−0.9	−0.8	−0.7
δ_{Rpi} (μm)	8.3	9.4	9.8	8.5	7.4	7.2	8.0	8.3	6.5	8.0	8.1
<i>i</i>	34	35	36	37	38	39	40	41	42	43	44
δ_{rpi} (μm)	−1.1	−1.1	−0.6	−0.9	−0.9	−0.4	−0.8	−0.9	−0.8	−0.7	−0.6
δ_{Rpi} (μm)	8.8	8.0	7.4	7.9	6.2	9.5	6.3	8.6	7.2	8.7	9.9
<i>i</i>	45	46	47	48	49	50	51	52			
δ_{rpi} (μm)	−0.7	−0.6	−0.6	−0.7	−0.8	−0.7	−0.6	−0.6			
δ_{Rpi} (μm)	7.8	8.4	9.0	7.7	7.8	8.3	9.2	6.9			

Table A2. Components measurement results of λ_{pi} and λ_{cj} .

i	1	2	3	4	5	6	7	8	9	10	11
λ_{pi} (arcsec)	-3.5	-0.2	-4.2	6.1	-1.8	0.3	0.2	-2.6	-2.9	1.8	-1.5
λ_{cj} (arcsec)	4.0	5.6	8.6	-2.2	1.5	0.9	-1.0	3.4	-0.5	-5.8	-4.4
i	12	13	14	15	16	17	18	19	20	21	22
λ_{pi} (arcsec)	3.5	-2.8	4.0	2.7	-8.0	1.5	-0.5	1.4	4.9	5.0	-1.5
λ_{cj} (arcsec)	0.8	-6.4	2.3	-3.6	2.2	-1.0	0.6	-4.0	6.5	2.0	1.6
i	23	24	25	26	27	28	29	30	31	32	33
λ_{pi} (arcsec)	0.3	-4.2	-2.3	1.5	-0.3	-1.3	3.2	-4.8	5.4	-0.6	-12.0
λ_{cj} (arcsec)	1.1	1.9	0.7	1.2	2.7	-2.8	6.4	-2.3	1.6	-5.6	-3.9
i	34	35	36	37	38	39	40	41	42	43	44
λ_{pi} (arcsec)	-0.5	-1.9	-2.3	-7.8	9.6	3.4	5	-5.1	0.1	4.0	4.4
λ_{cj} (arcsec)	0.1	-10.6	-5.4	2.7	1.3	0.4	8.3	1.7	1.2	8.9	-0.7
i	45	46	47	48	49	50	51	52			
λ_{pi} (arcsec)	-1.2	-2.7	-6.2	-0.5	0.6	-0.8	2.8	-4.0			
λ_{cj} (arcsec)	-1.3	2.4	-2.1	-0.6	2.1	8.3	10.6				

Table A3. Components measurement results of δ_{sk} , λ_{sk} and λ_{wk} .

k	1	2	3	4	5	6	7	8	9	10
δ_{sk} (μm)	1.6	0.5	1.1	0.7	1	0.9	1.2	1.3	1	0.8
λ_{wk} (arcsec)	5.5	4.9	-11.3	0.4	-1.7	-0.8	6.5	4.3	-1.1	-1.7
λ_{sk} (arcsec)	6.5	7.9	1.6	-3.8	-0.3	-1.5	1.4	8	3.2	-8.7